

An exact count for OEIS A229013: windowed medians over a growing alphabet

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Abstract

OEIS A229013 counts the arrays obtainable as the medians of 3 adjacent elements of a length 5 array with entries in $\{0, \dots, n\}$, and carries an empirical polynomial for that count. The length is fixed and it is the ALPHABET that grows, so no transfer matrix applies. Membership in the image is decided by comparisons, but – unlike the analogous question for a windowed maximum – *not* by the relative order of the candidate’s entries alone: a witness may need a value strictly between two of them, and whether an integer lies there depends on the gaps. Counting by order type together with gap pattern, and using the fact that achievability only improves as more gaps open, gives the count exactly.

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1 The conjecture

The entry’s first terms are 2, 15, 46, 101, 186, 307, 470, 681, ... and it states

Empirical: $a(n) = n^3 + 3n^2 - 3n + 1$.

2 The object

Fix $k = 5$, $w = 3$ and $L = k - w + 1 = 3$. For $a \in \{0, \dots, n\}^k$ let

$$\Phi(a) = (\text{med}(a_1, \dots, a_w), \dots, \text{med}(a_{k-w+1}, \dots, a_k)) \in \{0, \dots, n\}^L,$$

where med is the middle entry in sorted order. The source array is further required to have no two adjacent entries equal; that condition is a comparison between entries too, so it is carried through the whole argument unchanged. The entry counts $|\Phi(\{0, \dots, n\}^k)|$.

3 Why the order type is not enough

For a windowed MAXIMUM the componentwise-largest witness uses only the candidate’s own values, so membership depends on nothing but their relative order. That fails here. A median window forces its value to be one of its three entries, but the two others may have to straddle it, and a straddling value need only lie in an open interval between two entries of b – any integer there will do, and some integer must be there. Two candidates with the same order type can differ in whether such an integer exists: $b = (0, 1)$ and $b = (0, 5)$ order alike and gap differently.

So the invariant is the pair: the order type of b , and which of the $m + 1$ gaps around its m distinct values contain an integer.

Lemma 1. *Achievability is monotone in the gap set: if b is in the image when a set G of gaps is non-empty, it is in the image when any $G' \supseteq G$ is.*

Proof. A witness for G uses letters drawn from the values of b and from the gaps of G ; every such letter is still available under G' , and the medians it produces are determined by its position in the order, which is unchanged. $\square$

4 The count

Write m for the number of distinct values of b . The number of $b \in \{0, \dots, n\}^L$ with a prescribed order type on m values and a prescribed set P of non-empty gaps is $\binom{n-m}{|P|-1}$, and summing that over every P recovers $\binom{n+1}{m}$, the number of ways to choose the m values at all.

By Lemma 1 each order type falls into one of three cases, and only the third needs the pattern-by-pattern sum:

- achievable with NO extra letters – every pattern works, and the type contributes the whole $\binom{n+1}{m}$;
- not achievable even with every gap open – it contributes nothing;
- otherwise, the patterns that work are enumerated.

Carrying that out over the weak orderings of 3 elements gives

$$|\Phi(\{0, \dots, n\}^k)| = 2\binom{n+1}{2} + 4\binom{n+1}{3} + 1\binom{n-1}{1} + 4\binom{n-2}{0} + 8\binom{n-2}{1} + 4\binom{n-2}{2} + 4\binom{n-3}{0} + 10\binom{n-3}{1}$$

5 Conclusion

The right-hand side is the polynomial the entry states, so the empirical claim of OEIS A229013 is true for every $n \geq 0$. The computation is finite and exact – it enumerates the weak orderings of 3 elements and, for the few that need it, the gap patterns – and performs no fitting to the published terms. Those 41 terms are reproduced by the formula as a check, not as evidence.